BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 065 · Lesson 065 · Computational Methods: Recognising Solvable Problems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a precondition?

Q2. What is abstraction?

Q3. How does a queue differ from a stack?

Q4. 025

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

problem recognition — Identify the underlying task type.

constraint — Requirement a solution must obey.

objective function — Quantity to minimise or maximise.

computable specification — Precise task that a program can implement/check.

New learning

Identify — A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

Model — Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

Evaluate — Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

Worked example

Turn 'choose the best bus route' into a precise computational problem.

1. Explain the decision — Cost is defined and connections must be valid; output stops and total fare.

We Do · Practise with guidance

Turn 'pick the fairest lunch queue order' into a computable FIFO rule.

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Input ordered arrival timestamps and ticket IDs; output same order, oldest
  first; handle ties by sequence number.
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Explain the decision

An ordering rule makes fairness measurable and repeatable within the chosen model.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define computational problem input, process, output for cheapest route on a network. [6]

Question 2

A pupil asks to 'sort books by best'. Give two precise orderings, explain why different and select one for finding a title. [6]

Question 3

Evaluate whether a computer can choose a 'good teacher' simply from exam results, explaining models, constraints and limits. [6]

Exit Ticket · Answer before checking

Every vague human objective has exactly one obvious numerical optimisation target.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Condition required for an algorithm to work correctly.

Q2. Selecting relevant details and omitting irrelevant ones.

Q3. FIFO versus LIFO.

Q4. Binary search for 12 in sorted [2,4,6,8,10,12,14], show mid indexes.

You Do pages · Independent questions

Q1. Graph/start/goal [2], route algorithm evaluating costs [2], route and cost [2].

Q2. Alphabetical title [2], most-borrowed count [2], title lookup suits alphabetical ordering/search criterion [2].

Q3. Vague quality criterion [1], results limited proxy/confounding [2], relevant contextual inputs [1], risk unfair decisions [1], human judgement and wider evidence [1].

Exit Ticket · Model answer

False. Terms and constraints must be specified.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: